problem statement

Proposition 0.1. *Define $g_{n,i}(x) = I_x(i, n-i+1)$, we want to prove $g^{-1}_{n,n-j}(x)^n$ is an increasing function with respect to n, or $g_{n,n-j}(x^{\frac{1}{n}})$ is a decreasing function with respect to n. We have known $g_{n,n-j}(x)$ is a decreasing function with respect to n and $g^{-1}_{n,n-j}(x)$ is an increasing function with respect to n.*

I'm going to drop the $g_{i,n}$ notation here because it is very confusing. That notation should only be used when there are space constraints.

We will show that

$$I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) > 0.$$

Assume $x^{\frac{1}{n}} > j/n$:

Integration by parts allows us to write

$$\begin{aligned}
&I_{x^{\frac{1}{n}}}(n-j, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) \\
&\quad = I_{x^{\frac{1}{n}}}(n-j+1, j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1, n-j) + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(j+1)B(n-j, j+1)} \\
&\quad = -\int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1, j+1)}dy + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(j+1)B(n-j, j+1)}.
\end{aligned}$$

The function $y^a(1-y)^b$ achieves its maximum value at $y^* = a/(a+b)$. The assumption that $x^{\frac{1}{n}} > j/n$ implies that the function $y^{n-j}(1-y)^j$ is decreasing over $(x^{\frac{1}{n}}, x^{\frac{1}{n+1}})$. This implies that

$$\begin{aligned}
&-\int_{x^{\frac{1}{n}}}^{x^{\frac{1}{n+1}}} \frac{y^{n-j}(1-y)^j}{B(n-j+1, j+1)}dy + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(j+1)B(n-j, j+1)} \\
&\quad > \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}(x^{\frac{1}{n}} - x^{\frac{1}{n+1}})}{B(n-j+1, j+1)} + \frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j+1}}{(j+1)B(n-j, j+1)}.
\end{aligned}$$

And $B(n-j+1,j+1) = B(n-j,j+1)\frac{j+1}{n+1}$ allows us to simplify the right hand side of the above inequality to

$$\frac{\left(x^{\frac{1}{n}}\right)^{n-j}\left(1-x^{\frac{1}{n}}\right)^{j}}{(j+1)B(n-j,j+1)}\left(1-x^{\frac{1}{n}}+(n+1)(x^{\frac{1}{n}}-x^{\frac{1}{n+1}})\right).$$

Note that

$$1-x^{\frac{1}{n}}+(n+1)(x^{\frac{1}{n}}-x^{\frac{1}{n+1}}) = 1+nx^{\frac{1}{n}}-(n+1)x^{\frac{1}{n+1}} > 0$$

for all $x \in (0,1)$. Putting it all together, we have

$$I_{x^{\frac{1}{n}}}(n-j,j+1) - I_{x^{\frac{1}{n+1}}}(n-j+1,n-j) > 0,$$

when $x^{\frac{1}{n}} > j/n$.